Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Luis A. Pimentel Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
11375 Bruceville RD	Elk Grove	Sacramento	95758

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1953

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X132-X240-X007-XXXX

B. OPERATORS

Pimentel, Luis

Operator name: Pimentel, Luis

Telephone no.: Landline: ; Cellular: (123) 000-0000

Mailing Address Number and Street	City	State	Zip Code
11375 Bruceville RD	Elk Grove	CA	95758

This operator is responsible for paying permit fees.

C. OWNERS

Pimentel, Edward & Luis Trust

Legal owner name: Pimentel, Edward & Luis Trust

Telephone no.: Landline: ; Cellular: (123) 000-0000

Mailing Address Number and Street	City	State	Zip Code
11375 Bruceville RD	Elk Grove	CA	95758

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	35	23	81	59	0
Number under roof	260	0	0	0	0	0
Maximum number	260	35	23	81	59	0
Average number	260	35	23	81	59	0
Avg live weight (lbs)	1,350	1,350	1,150	400		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 8,248.74 *tons per reporting period*

Total nitrogen from manure: 105,672.91 *lbs per reporting period*

After ammonia losses (30% loss applied): 73,971.04 *lbs per reporting period*

Total phosphorus from manure: 17,342.11 *lbs per reporting period*

Total potassium from manure: 49,450.76 *lbs per reporting period*

Total salt from manure: 130,469.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,908,000 *gallons*

Total nitrogen generated: 9,395.34 *lbs*

Total phosphorus generated: 1,935.17 *lbs*

Total potassium generated: 16,090.29 *lbs*

Total salt generated: 105,337.27 *lbs*

	4,908,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,908,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #3	Ground water
Ag Well #5	Ground water
Domestic Well Home Pace Maine	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/30/2023	Liquid commercial fertilizer *82-0-0 NH3*	1,680.00 *gal*			82.000000	0.000000	0.000000	0.000000
06/01/2023	Liquid commercial fertilizer *4-10-10*	1,332.40 *gal*			4.000000	10.000000	10.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	11,940.83	1,111.89	1,111.89	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	11,940.83	1,111.89	1,111.89	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1A	40	40	2	process wastewater	X132-X240-X007-XXXX X132-X240-X064-XXXX
1B	40	40	0	none	X132-X240-X007-XXXX X132-X240-X064-XXXX
2	20	20	2	process wastewater	X132-X240-X007-XXXX
7	20	20	0	none	X132-X240-X015-XXXX
8	20	20	0	none	X132-X310-X022-XXXX
9	20	20	0	none	X132-X310-X022-XXXX
Totals for areas that were used for application	60	60	4		
Totals for areas that were not used for application	100	100	0		
Land application area totals	160	160	4		

B. CROPS AND HARVESTS

1A

Field name: 1A

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	640.00 *ton*	Dry-weight		88.7	16,700.00	1,820.00	13,700.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	60.39	6.58	49.54	408.61

1A

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	1,200.00 *ton*	Dry-weight		64.9	9,760.00	2,150.00	11,000.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	205.55	45.28	231.66	1,305.72

2

Field name: 2

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	320.00 *ton*	Dry-weight		63.9	18,200.00	2,040.00	13,400.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	210.25	23.57	154.80	1,189.86

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/30/2023	600.00 *ton*	Dry-weight		62.8	9,050.00	1,640.00	6,250.00		7.36

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	202.00	36.60	139.50	1,642.75

NUTRIENT BUDGET

A. LAND APPLICATIONS

1A - 11/05/2022: Oats, silage-soft dough

Field name: 1A

Crop: Oats, silage-soft dough — Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	27.58	5.70	43.98	243.34	432,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	109.35	2,016,000.00 *gal*
Application event totals		27.58	5.70	43.98	352.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.84	0.53	4.47	232.83	360,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	109.35	2,016,000.00 *gal*
Application event totals		28.84	0.53	4.47	342.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	17.30	0.32	2.68	139.70	216,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	109.35	2,016,000.00 *gal*
Application event totals		17.30	0.32	2.68	249.05	

1A - 06/07/2023: Corn, silage

Field name: 1A

Crop: Corn, silage — Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1A - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
82-0-0	Liquid commercial fertilizer	177.00	0.00	0.00	0.00	
Application event totals		177.00	0.00	0.00	0.00	

06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	246.04	4,536,000.00 *gal*
Application event totals		0.00	0.00	0.00	246.04	

06/05/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Liquid commercial fertilizer	13.90	34.50	34.50	0.00	
Application event totals		13.90	34.50	34.50	0.00	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	246.04	4,536,000.00 *gal*
Application event totals		0.00	0.00	0.00	246.04	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.10	13.22	111.76	576.81	1,152,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	164.03	3,024,000.00 *gal*
Application event totals		41.10	13.22	111.76	740.84	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	195.92	3,612,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.92	

1A - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	246.04	4,536,000.00 *gal*
Application event totals		0.00	0.00	0.00	246.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.10	13.22	111.76	576.81	1,152,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	195.92	3,612,000.00 *gal*
Application event totals		41.10	13.22	111.76	772.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	246.04	4,536,000.00 *gal*
Application event totals		0.00	0.00	0.00	246.04	

2 - 11/05/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.17	6.66	51.31	283.90	252,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	136.69	1,260,000.00 *gal*
Application event totals		32.17	6.66	51.31	420.59	

2 - 11/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.65	0.62	5.21	271.63	210,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	136.69	1,260,000.00 *gal*
Application event totals		33.65	0.62	5.21	408.32	

04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	20.19	0.37	3.13	162.98	126,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	136.69	1,260,000.00 *gal*
Application event totals		20.19	0.37	3.13	299.67	

2 - 06/07/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
82-0-0 NH3	Liquid commercial fertilizer	177.00	0.00	0.00	0.00	
Application event totals		177.00	0.00	0.00	0.00	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	328.06	3,024,000.00 *gal*
Application event totals		0.00	0.00	0.00	328.06	

2 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Liquid commercial fertilizer	13.90	34.50	34.50	0.00	
Application event totals		13.90	34.50	34.50	0.00	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	328.06	3,024,000.00 *gal*
Application event totals		0.00	0.00	0.00	328.06	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	41.10	13.22	111.76	576.81	576,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	273.38	2,520,000.00 *gal*
Application event totals		41.10	13.22	111.76	850.19	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	273.38	2,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	273.38	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	218.71	2,016,000.00 *gal*
Application event totals		0.00	0.00	0.00	218.71	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.82	9.91	83.82	432.60	432,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	218.71	2,016,000.00 *gal*
Application event totals		30.82	9.91	83.82	651.31	

2 - 06/07/2023: Corn, silage

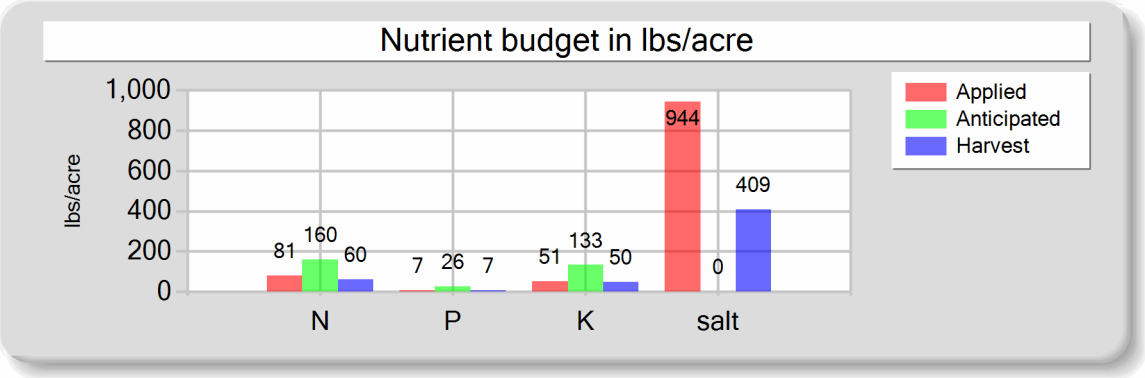
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	164.03	1,512,000.00 *gal*
Application event totals		0.00	0.00	0.00	164.03	

B. NUTRIENT BUDGET

1A - 11/05/2022: Oats, silage-soft dough

Field name: 1A | Crop: Oats, silage-soft dough | Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	73.72	6.55	51.13	615.86
Fresh water	0.00	0.00	0.00	328.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	80.72	6.55	51.13	943.92
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	60.39	6.58	49.54	408.61
Nutrient balance	20.34	-0.03	1.59	535.31
Applied to removed ratio	1.34	1.00	1.03	2.31

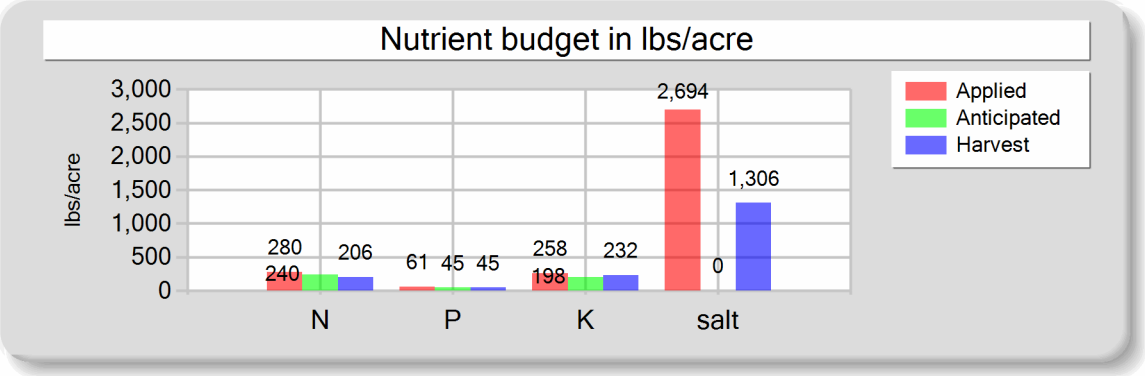
Fresh water applied	
6,048,000.00	*gallons*
222.73	*acre-inches*
5.57	*inches/acre*

Process wastewater applied	
1,008,000.00	*gallons*
37.12	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

1A - 06/07/2023: Corn, silage

Field name: 1A | Crop: Corn, silage | Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	190.90	34.50	34.50	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	82.19	26.44	223.51	1,153.61
Fresh water	0.00	0.00	0.00	1,540.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.09	60.94	258.01	2,693.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	205.55	45.28	231.66	1,305.72
Nutrient balance	74.55	15.66	26.35	1,387.95
Applied to removed ratio	1.36	1.35	1.11	2.06

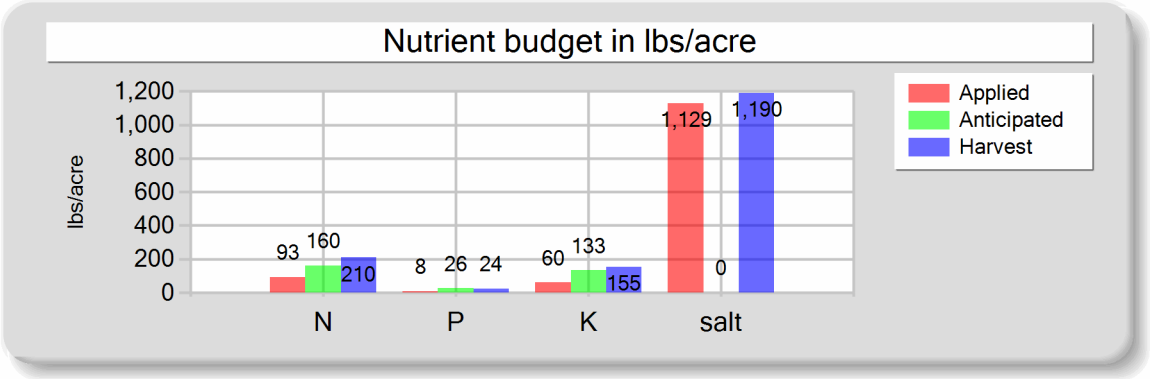
Fresh water applied	
28,392,000.00	*gallons*
1,045.58	*acre-inches*
26.14	*inches/acre*

Process wastewater applied	
2,304,000.00	*gallons*
84.85	*acre-inches*
2.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/05/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	86.01	7.65	59.65	718.50
Fresh water	0.00	0.00	0.00	410.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	93.01	7.65	59.65	1,128.58
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	210.25	23.57	154.80	1,189.86
Nutrient balance	-117.24	-15.92	-95.14	-61.28
Applied to removed ratio	0.44	0.32	0.39	0.95

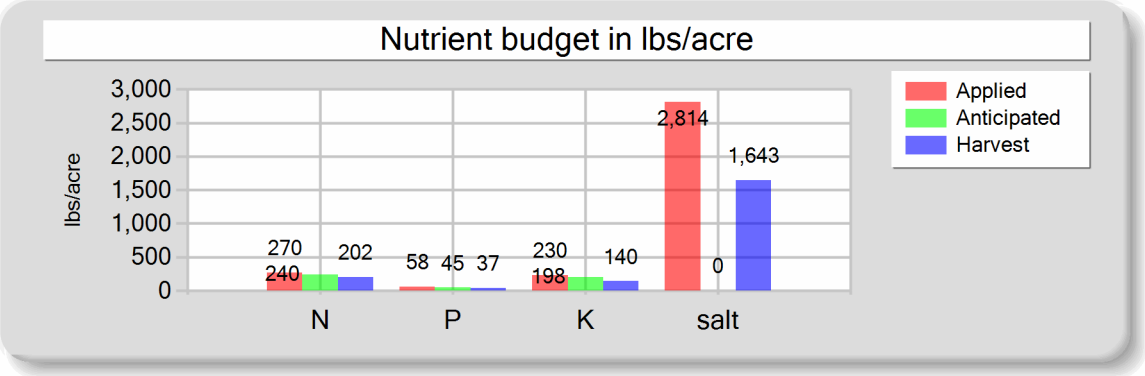
Fresh water applied	
3,780,000.00	*gallons*
139.20	*acre-inches*
6.96	*inches/acre*

Process wastewater applied	
588,000.00	*gallons*
21.65	*acre-inches*
1.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/07/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	190.90	34.50	34.50	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	71.92	23.13	195.57	1,009.41
Fresh water	0.00	0.00	0.00	1,804.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.82	57.63	230.07	2,813.73
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	202.00	36.60	139.50	1,642.75
Nutrient balance	67.82	21.03	90.57	1,170.98
Applied to removed ratio	1.34	1.57	1.65	1.71

Fresh water applied	
16,632,000.00	*gallons*
612.50	*acre-inches*
30.63	*inches/acre*

Process wastewater applied	
1,008,000.00	*gallons*
37.12	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2102607-01

Sample and source description: Manure 2102607-01

Sample date: 10/26/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,140.00	7,360.00	42,200.00	12,000.00	11,000.00	5,600.00	738.00	12,900.00		20.40
DL	0.50	0.50	5.00	5.00	5.00	5.00	1.00	0.50		0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2301425-01

Sample and source description: Lagoon Water 2301425-01

Sample date: 01/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	306.00	142.00			63.30	488.00								3,360.00	2,700
DL	0.60	6.00			0.10	10.00								1.00	1

Lagoon Water 2303488-01

Sample and source description: Lagoon Water 2303488-01

Sample date: 03/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	384.00	22.30			7.06	59.50								4,110.00	3,100
DL	0.60	0.60			0.10	1.00								1.00	1

Lagoon Water 2307291-01

Sample and source description: Lagoon Water 2307291-01

Sample date: 07/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	171.00	171.00			55.00	465.00								3,460.00	2,400
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310011-01

Sample and source description: Lagoon Water 2310011-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	131.00	70.60			40.60	453.00								2,950.00	2,200
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well #3

2401055-02

Sample description: 2401055-02

Sample date: 01/04/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											281.00	260
DL											1.00	1

Ag Well #5

Ag Well #5

2401055-03

Sample description: 2401055-03

Sample date: 01/04/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											240.00	1,500
DL											1.00	1

Domestic Well Home Pace Maine

2401055-01

Sample description: 2401055-01

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			7.09								623.00	
DL			0.40								1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1A - 11/05/2022: Oats, silage-soft dough

1A - 11/05/2022: Oats, silage-soft dough

2305222-01

Sample and source description: 2305222-01

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 88.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	1,820.00	13,700.00		11.30
DL	0.50	0.50	0.50		0.10

1A - 06/07/2023: Corn, silage

2311036-01

Sample and source description: 2311036-01

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,760.00	2,150.00	11,000.00		6.20
DL	0.50	0.50	0.50		0.10

2 - 11/05/2022: Oats, silage-soft dough

2305222-02

Sample and source description: 2305222-02

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	2,040.00	13,400.00		10.30
DL	2.50	0.50	0.50		0.10

2 - 06/07/2023: Corn, silage

2311036-02

Sample and source description: 2311036-02

Sample date: 11/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,050.00	1,640.00	6,250.00		7.36
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

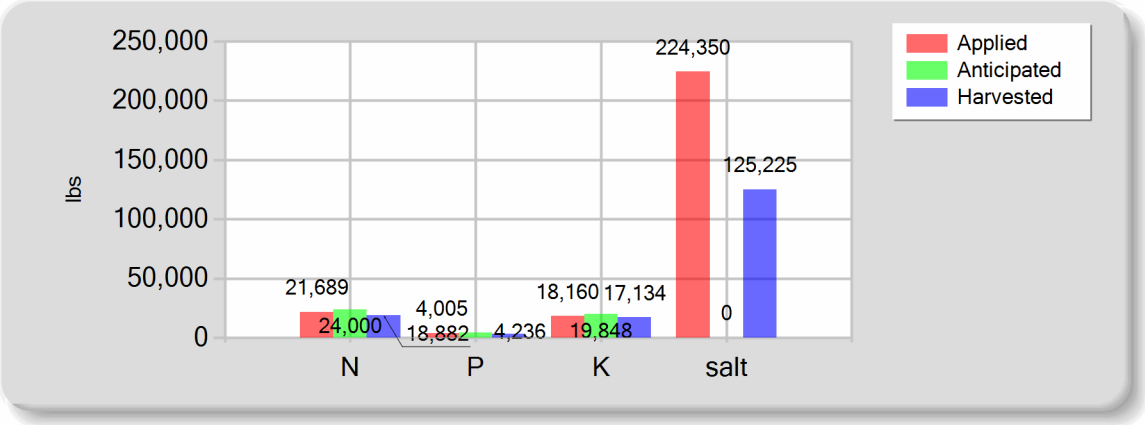
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

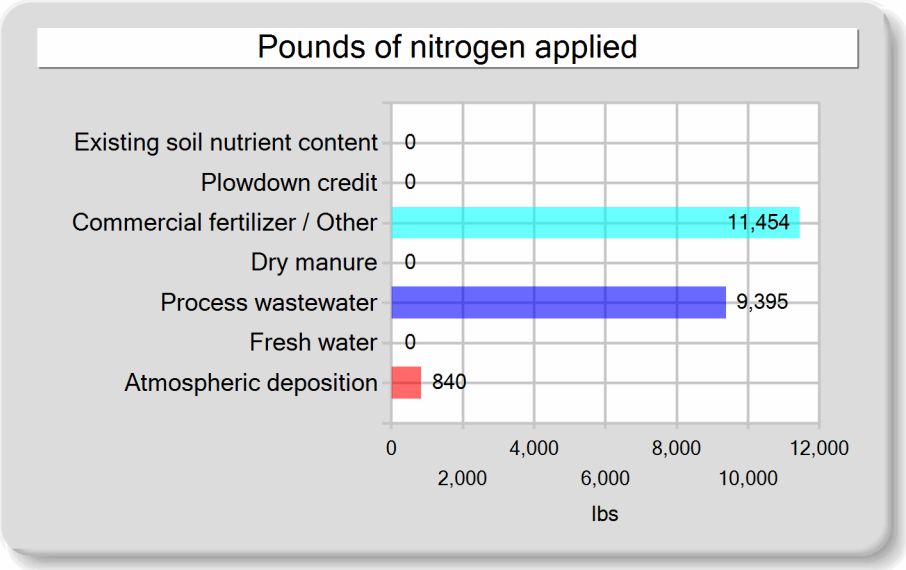
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

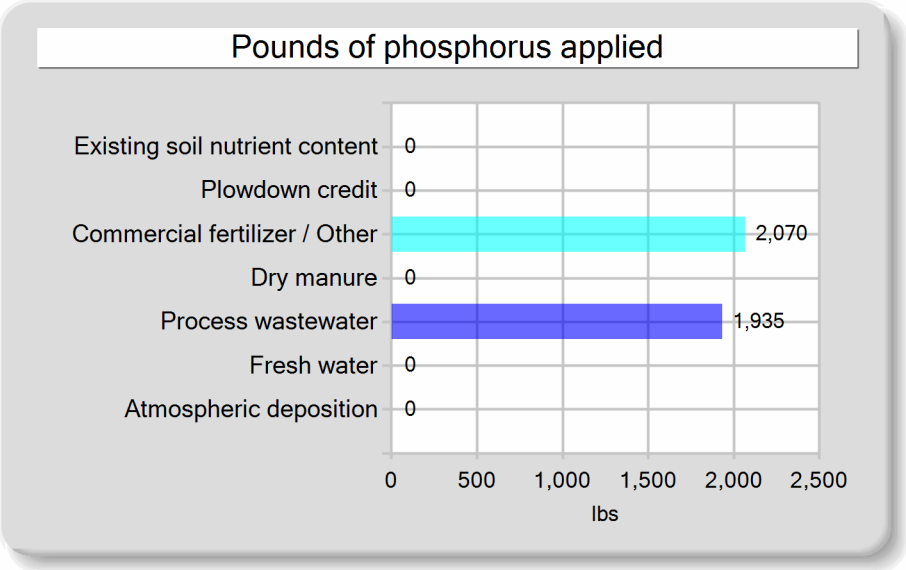
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	11,454.00	2,070.00	2,070.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	9,395.34	1,935.17	16,090.29	105,337.27
Fresh water	0.00	0.00	0.00	119,012.38
Atmospheric deposition	840.00	0.00	0.00	0.00
Total nutrients applied	21,689.34	4,005.17	18,160.29	224,349.65
Anticipated crop nutrient removal	24,000.00	4,236.00	19,848.00	0.00
Actual crop nutrient removal	18,882.16	3,277.82	17,133.90	125,225.28
Nutrient balance	2,807.18	727.35	1,026.39	99,124.37
Applied to removed ratio	1.15	1.22	1.06	1.79

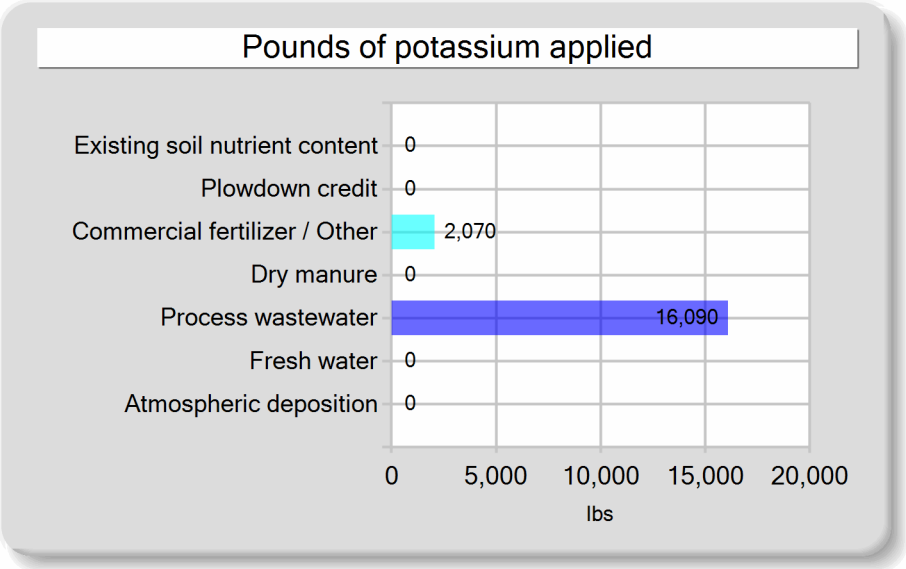
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

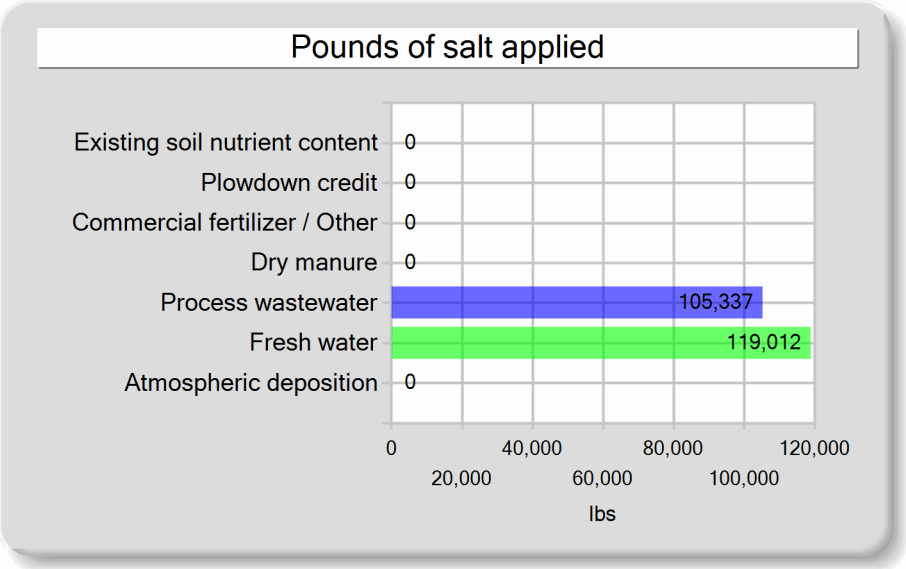


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Edward & Luis Trust Pimentel	Luis Pimentel
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Edward Pimentel	Luis A. Pimentel
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	N/A
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	